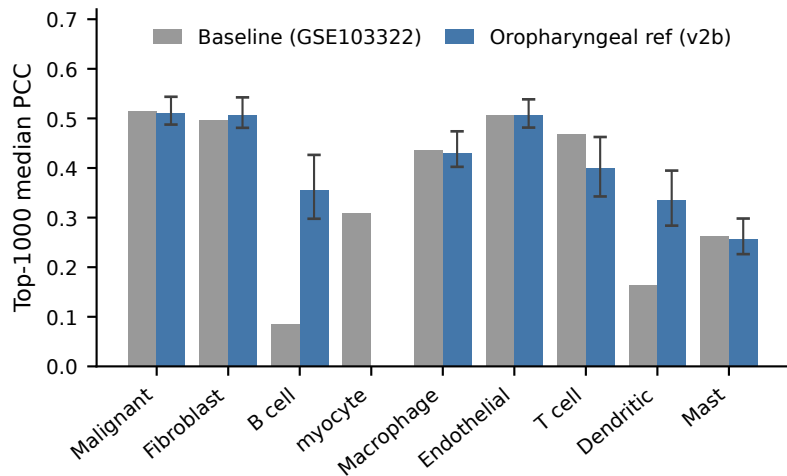


Single-cell reference choice / B-cell repair

a. Reference swap rescues B-cell / dendritic prediction



error bars: 95% CI, patient bootstrap (1000 resamples; shown for the oropharyngeal reference only, the cached baseline stores one scalar per cell type)

Reference construction (B-cell repair)

- Baseline GSE103322 atlas has only 138 B cells → BayesPrism deconvolves B cells to ~0% in 93% of TCGA tumors → labels near-zero → B-cell prediction collapses (PCC 0.085, 0 genes > 0.4).
- Fix: oropharyngeal HPV-mixed atlas (3332 B cells) + relaxed gene filter (nonzero ≥ 0.2). B cell 0.085→0.34, Dendritic 0.16→0.35; T cell small accepted cost 0.47→0.40.
- Single reference used uniformly for TCGA and CPTAC (no per-cell-type cherry-picking).